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✉ For correspondence:

afenton@nyu.edu

tsacktor@downstate.edu

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* Present address: Department of Neuroscience and Cell Biology, Rutgers University, New Brunswick, United States

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PKM ζ -PKC ι/λ double-knockout demonstrates atypical PKC is crucial for the persistence of hippocampal LTP and spatial memory

Panayiotis Tsokas^{1,2,3}, Changchi Hsieh¹, Alejandro Grau-Perales⁴, Andrew Tcherepanov¹, Leo Kwok⁴, Laura Melissa Rodriguez-Valencia^{1,2}, David A Cano⁵, Kim D Allen^{1,3,6}, Hannah JH Smith^{1,*}, Sabina Kubayeva⁵, Benson J Wei¹, Samuel Sabzanov⁵, Rafael E Flores-Obando¹, Sourav Ghosh⁷, Peter John Bergold^{1,8}, Jerry Rudy⁹, James E Cottrell², André Antonio Fenton^{1,4,10}✉, Todd Charlton Sacktor^{1,2,8}✉

¹Department of Physiology and Pharmacology, The Robert F. Furchgott Center for Neural and Behavioral Science, State University of New York Downstate Health Sciences University, Brooklyn, United States • ²Department of Anesthesiology, State University of New York Downstate Health Sciences University, Brooklyn, United States • ³Department of Pathology, State University of New York Downstate Health Sciences University, Brooklyn, United States • ⁴Center for Neural Science, New York University, New York, United States • ⁵College of Medicine, State University of New York Downstate Health Sciences University, Brooklyn, United States • ⁶Department of Biology, City University of New York-Medgar Evers College, Brooklyn, United States • ⁷Departments of Neurology and Pharmacology, Yale University, New Haven, United States • ⁸Department of Neurology, State University of New York Downstate Health Sciences University, Brooklyn, United States • ⁹Department of Psychology and Neuroscience, University of Colorado at Boulder, Boulder, United States • ¹⁰Neuroscience Institute at NYU Langone Medical Center, New York, United States

eLife Assessment

This **important** study addresses the unresolved and long-debated question of whether atypical protein kinase C is required for the maintenance of synaptic potentiation and long-term memory. The results confirm previous findings that persistent activity of PKM ζ is required for lasting potentiation of hippocampal synapses and spatial memory. The study also adds new genetic evidence to support the earlier suggestion that enhanced expression of PKC ι/λ compensates for the genetic reduction of PKM ζ to support synaptic potentiation and memory; however, the results as currently presented were viewed as **incomplete**.

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Abstract

Is long-term memory maintained by stable synaptic structures or persistent biochemical action? PKM ζ , a persistently active atypical PKC (aPKC), is thought to be crucial for maintaining the late phase of long-term potentiation (late-LTP) and long-term memory through sustained kinase action. However, PKM ζ -knockout mice express hippocampal LTP and spatial memory while lacking neocortical LTP, calling into question whether persistent kinases and PKM ζ are fundamental to LTP and memory. In Tsokas *et al.*, 2016, we showed PKC ι/λ , the other aPKC, is the most likely compensating PKC isoform in PKM ζ -knockout mice. In wild-type mice, PKC ι/λ is critical to the initial generation of early-LTP and short-term memory, and PKM ζ compensates for genetic deletion of PKC ι/λ by supporting both early- and late-phase processes. Here, we found PKC ι/λ persists in LTP and long-term memory when PKM ζ is genetically deleted. We tested whether PKC ι/λ

λ functionally compensates for the loss of PKM ζ by genetically ablating both aPKCs. Whereas deleting PKM ζ and PKC ι/λ individually induces compensation, deleting both aPKCs abolishes hippocampal late-LTP. Hippocampal ι/λ - ζ -double-knockout eliminates spatial long-term memory but not short-term memory. Thus, in the absence of PKM ζ , a second persistent biochemical process compensates to maintain late-LTP and long-term memory.

Introduction

The prevailing textbook mechanism for how memory is retained asserts that stable structural changes at synapses, the result of initial protein synthesis and growth, sustain memory without the need for ongoing biochemical activity dedicated to storing information. This idea, as introduced, hypothesized, and established by Ramón y Cajal, Hebb, and Kandel, has led to identifying structural plasticity, and structural components of synapses that contribute to establishing LTP and memory (Ramón y Cajal, 1894 [\[1\]](#); Hebb, 1949 [\[2\]](#); Bailey and Kandel, 1993 [\[3\]](#)), including cytoskeletal actin and perineuronal nets (Matus, 2000 [\[4\]](#); Tsien, 2013 [\[5\]](#)). In contrast to non-enzymatic, stable changes in structure, memory could also be sustained by ongoing, persistent biochemical processes, a key alternative hypothesis independently proposed by Crick, Lisman, and Schwartz (Crick, 1984 [\[6\]](#); Lisman, 1985 [\[7\]](#); Schwartz, 1993 [\[8\]](#)).

The search for biochemical processes that maintain long-term potentiation (LTP) for hours and long-term memory for days has focused on the persistently active protein kinases, CaMKII and PKM ζ . The kinase action of CaMKII is only crucial for initiating but not perpetuating LTP and memory (Sacktor and Fenton, 2018 [\[9\]](#); Tullis et al., 2023 [\[10\]](#); Bayer and Giese, 2024 [\[11\]](#)). In contrast, under normal circumstances, the autonomously active, atypical PKC (aPKC) isoform PKM ζ is thought to play a critical role in maintaining the enduring late phase of LTP and long-term memory (Pastalkova et al., 2006 [\[12\]](#); Shema et al., 2007 [\[13\]](#); Shema et al., 2011 [\[14\]](#); Wang et al., 2016 [\[15\]](#); Tsokas et al., 2024 [\[16\]](#)). Application of zeta-inhibitory peptide (ZIP) that suppresses both PKM ζ and PKC ι/λ , a closely related aPKC, reverses late-LTP, even after a day (Pastalkova et al., 2006 [\[12\]](#); Madronal et al., 2010 [\[17\]](#)). (Hereafter PKC ι/λ is referred to as PKC ι .) ZIP also erases enduring long-term memory for specific but not general types of information, days (Serrano et al., 2008 [\[18\]](#); Gao et al., 2018 [\[19\]](#)) and even months (Pastalkova et al., 2006 [\[12\]](#); Shema et al., 2007 [\[13\]](#)) after memory acquisition without impairing new learning. There is concern that the inhibitory effects of ZIP on LTP and memory maintenance may not be due to its targeting PKM ζ or PKC ι , but to its off-target effects (Sadeh et al., 2015 [\[20\]](#); LeBlancq et al., 2016 [\[21\]](#); Stokes et al., 2025 [\[22\]](#)). Nonetheless, multiple second-generation PKM ζ -specific inhibitors, PKM ζ -antisense oligodeoxynucleotides, and PKM ζ -shRNA also disrupt late-LTP and long-term memory (Tsokas et al., 2016 [\[23\]](#); Wang et al., 2016 [\[15\]](#); Tsokas et al., 2024 [\[16\]](#)).

If PKM ζ is genetically deleted, however, the key long-term processes of hippocampal LTP and long-term memory appear normal and are still inhibited by ZIP (Lee et al., 2013 [\[24\]](#); Volk et al., 2013 [\[25\]](#); Tsokas et al., 2016 [\[23\]](#)). In contrast, in the same mutant mice LTP is eliminated in medial prefrontal cortex (mPFC) (Kniffin et al., 2025 [\[26\]](#); Sacktor, 2025 [\[27\]](#)). These results appear contradictory, suggesting that PKM ζ alone may be essential for LTP maintenance in mPFC, whereas either other ZIP-sensitive molecules compensate for PKM ζ in the hippocampus, or an enduring structural change sustains LTP and memory in the absence of *Prkcz*, the PKM ζ gene. In Tsokas et al., 2016 [\[23\]](#), several members of the PKC gene family were found to increase expression in compensation for the loss of PKM ζ , and PKC ι was the most promising candidate, as closely related genes often functionally compensate (Gu et al., 2003 [\[28\]](#); Conant and Wagner, 2004 [\[29\]](#); White et al., 2013 [\[30\]](#); El-Brolosy and Stainier, 2017 [\[31\]](#)). Notably, PKC ι is critical to the initial generation of the early phase of LTP and short-term memory (Ren et al., 2013 [\[32\]](#); Tsokas et al., 2016 [\[23\]](#); Wang et al., 2016 [\[15\]](#); Sheng et al., 2017 [\[33\]](#)), and PKM ζ compensates for its genetic deletion, supporting both short- and long-term processes (Sheng et al., 2017 [\[33\]](#)). Perhaps instead of a structural change from activity-dependent growth, LTP and long-term memory maintenance might require a persistent kinase after all — but not always PKM ζ .

The notion that persistent PKC ι is a biochemical mechanism responsible for late-phase LTP and long-term memory when PKM ζ is genetically deleted raises two questions: 1) Does PKC ι persist in LTP and long-term memory in the absence of PKM ζ , and 2) Does eliminating the two biochemical mechanisms abolish enduring LTP and long-term memory? We address these questions using conditional and double-knockout strategies.

Results

As neither inducible nor constitutive PKM ζ gene ablation eliminates hippocampal LTP (Volk et al., 2013), it is important to know if compensatory increases in other PKCs are present in the hippocampus of conditional PKM ζ -KO mice (ζ -cKO), as observed in PKM ζ -null mice (Tsokas et al., 2016). In PKM ζ -null mouse hippocampus, basal levels of PKC ι and the conventional PKC β I specifically increased (Tsokas et al., 2016). To examine ζ -cKO mouse hippocampus, adult CaMKII α promoter-driven-CreER^{T2}Prkcz^{fl/fl} mice were injected with tamoxifen. One week later, expression of PKM ζ in hippocampus decreased, and there was a compensatory increase in PKC ι , as well as PKC ι phosphorylated on its activation-loop (Figure 1A, Figure 1 — figure supplement 1A, Table S1A, 2A). In addition, the expression of all four conventional PKCs (α , β I, β II, γ) increased, as did phosphorylation of the conventional PKC activation-loop (Figure 1B, Figure 1 — figure supplement 1A, Table S1B, 2A). However, there was no change in the expression of the four novel PKC isoforms (δ , ϵ , η , θ) or phosphorylation of the activation-loop of PKC ϵ (Figure 1B, Figure 1 — figure supplement 1A, Table S1B, 2A). There also was no change in either the level of CaMKII α , which is critical for LTP induction (Bayer and Giese, 2024), or T286-autophosphorylation of CaMKII α , which induces Ca²⁺-independent autonomous kinase activity (Miller and Kennedy, 1986) (Figure 1 — figure supplement 1B, Table S2B). Thus, conditional and constitutive PKM ζ -KO mice express similar compensatory increases in PKC ι , as well as other PKCs, in hippocampus.

Does the increase in PKC ι persist in long-term memory, such that one biochemical process substitutes for the other? Previous research has shown that PKM ζ expression in CA1 *stratum* (*st.*) *radiatum* of wild-type (WT) mice remains elevated for at least a month after training on a spatial memory task (Hsieh et al., 2021; Han et al., 2025). We determined if PKM ζ -cKO mice would also exhibit a compensatory increase of PKC ι in *st. radiatum*. PKM ζ was ablated in CaMKII α -promoter-driven-CreER^{T2}Prkcz^{fl/fl} mice. Three weeks later, they were trained on the active place avoidance task (Figure 1C, Figure 1 — figure supplement 2). One week after training, compared to vehicle-injected littermates, the PKM ζ -cKO showed a decrease to ~25% in PKM ζ expression and an increase to ~400% in PKC ι expression (Figure 1C, Table S1C). Thus, ζ -cKO induces compensatory persistently increased expression of PKC ι .

Does the persistent PKC ι mechanism functionally substitute for that of PKM ζ in LTP and long-term memory of PKM ζ -KO mice? As PKC ι -null mice are embryonically lethal (Seidl et al., 2013), we determined the functional significance of the compensatory increase of PKC ι for LTP by injecting an adeno-associated virus (AAV) expressing Cre-recombinase in one hippocampus of PKC ι -floxed/PKM ζ -null mice (*Prkci*^{fl/fl}-*Prkcz*^{-/-}) mice (Sheng et al., 2017) (Figure 2). The contralateral hippocampus was injected with a control AAV-eGFP, and *ex vivo* slices prepared for electrophysiology. PKC ι decreased in the ipsilateral hippocampus to ~20% compared to the contralateral hippocampus (Figure 2). If PKC ι is important for enduring LTP in PKM ζ -KO mice, then this decrease should disrupt LTP. High-frequency stimulation (HFS) of the Schaffer collateral/commissural-CA1 synapses in the ipsilateral ι -dKO slices produced no persistent change in the residual PKC ι and a transient LTP only lasting ~1-2 h (Figures 2, 3A). In contrast, HFS of the contralateral slices with PKC ι but lacking PKM ζ induced persistent increases of the PKC ι to ~200% and late-LTP, both lasting at least 3 h, the duration of the recordings (Figures 2 and 3A). In addition, slices from hippocampus of PKC ι -floxed/PKM ζ -null mice injected with AAV expressing Cre by the CaMKII α -promoter to selectively delete PKC ι in excitatory neurons resulted in similar transient LTP (Figure 3A, inset; Figure 3 — figure supplement 1A). The virus injected into PKC ι -floxed mice that express PKM ζ (*Prkci*^{fl/fl}-*Prkc*^{+/+} mice) resulted in

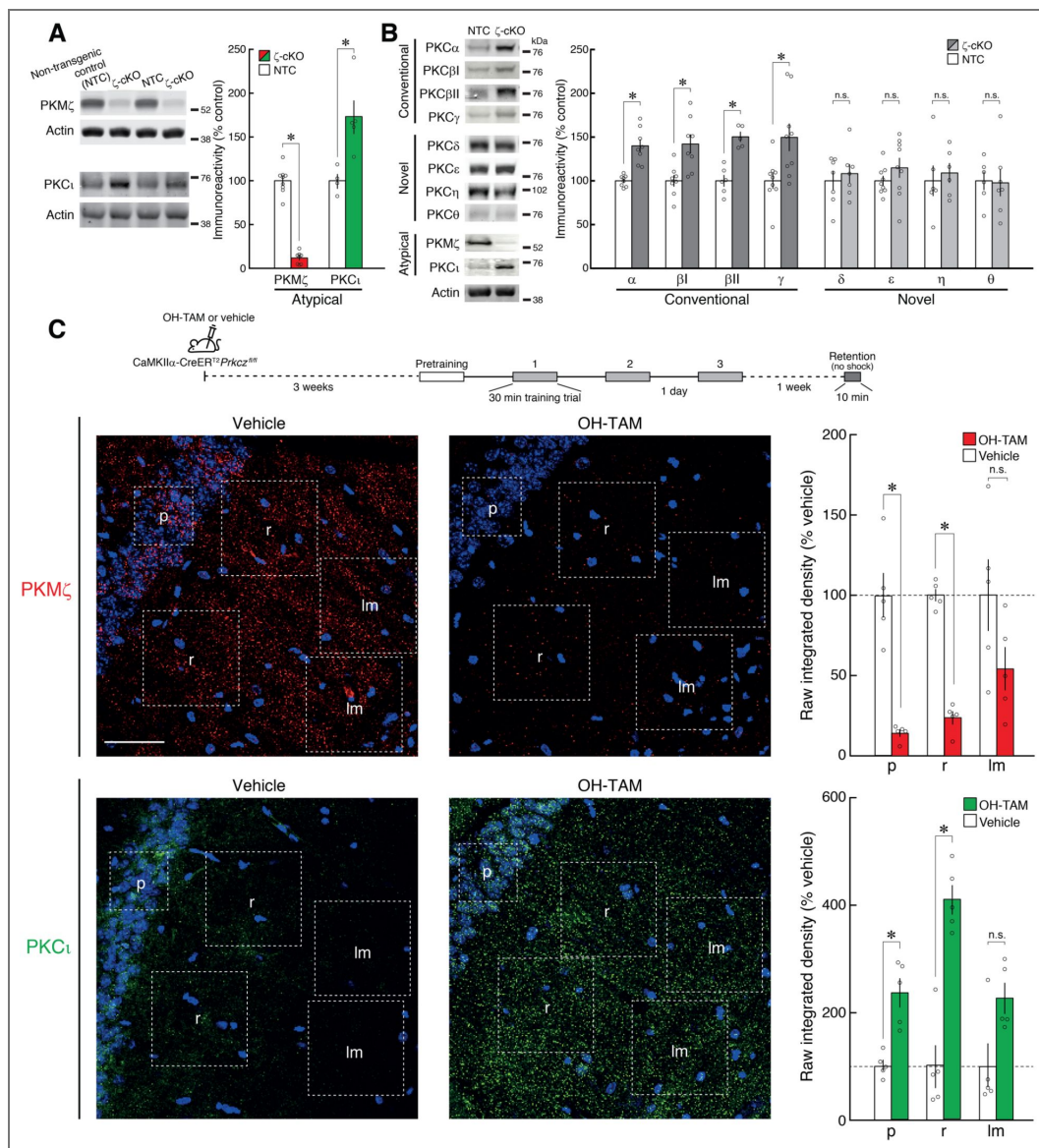


Figure 1. Compensatory increases of atypical PKC ι and conventional, but not novel PKCs, in conditional PKM ζ -KO mouse hippocampus.

(A, B) Immunoblots of hippocampal extracts from *Camk2a-CreER^{T2}Prkcz^{fl/fl}* mice receiving tamoxifen (2 mg/200 μ l i.p., 5 daily doses) to activate Cre recombinase selectively in excitatory neurons. Mice are sacrificed 7 days after the last dose. Left, representative immunoblots with Mr markers. Right, mean $\pm$ SEM. Significance by two sample Student *t* tests with Bonferroni correction denoted by *; not significant, n.s.; statistics in Table S1A,B. Tamoxifen is a partial PKC antagonist and may still be present after a week (O'Brian et al., 1985); therefore, WT mice that also received tamoxifen are non-transgenic controls (NTC). (A) PKM ζ decreases and PKC ι increases in PKM ζ -cKO mice. (B) Conventional PKCs increase and novel PKCs do not change. For clarity, the actin shown is for PKCs α , ι , and PKM ζ immunoblots. (C) Immunohistochemistry shows ζ -cKO reduces PKM ζ and increases PKC ι in CA1 *st. pyramidale* (p) and *radiatum* (r), but not *lacunosum-moleculare* (lm) 1 week after training. Inset above, schematic of experimental protocol. PKM ζ is deleted using *Camk2a-CreER^{T2}Prkcz^{fl/fl}* mice. Cre is activated using 4-OH tamoxifen (OH-TAM, 2 mg/200 μ l i.p., 3 doses every other day). Control mice receive vehicle injections. Active place avoidance training begins 3 weeks later, and 1 week after training memory retention is tested in the absence of shock. Left above, immunohistochemistry reveals reduced PKM ζ expression in cell bodies and dendritic compartments of the PKM ζ -cKO. Left below, PKC ι increases in cell bodies as well as in dendritic compartments where it is ordinarily expressed at low levels. Right, mean $\pm$ SEM. Student *t* tests with Bonferroni corrections compared differences in PKM ζ and PKC ι expression separately in strata of CA1 (Table S1C). DAPI staining of nuclei shown in blue. Bar = 50 μ m.

compensated LTP as in previous reports (Sheng et al., 2017) (Figure 3 — figure supplement 1B). Thus, deleting the aPKCs individually produces compensation, whereas deleting both aPKCs eliminates late-LTP.

We tested if late-LTP could be induced in ι/ζ -dKO hippocampus by increasing HFS from two trains, 20 sec apart, which is optimized to produce an early onset of late-LTP (Tsokas et al., 2007), to four trains, spaced 5 min apart, which is optimized to produce maximal late-LTP (Scharf et al., 2002; Serrano et al., 2005) (Figure 3B). Stronger stimulation induces LTP in ι/ζ -dKO slices that lasts only ~1-2 h. This LTP was no longer expressed at 3 h post-stimulation, at which time field excitatory postsynaptic potentials (fEPSPs) did not significantly differ from baseline fEPSPs before HFS.

To determine if PKC ι supports long-term memory in the absence of PKM ζ , we trained mice on the active place avoidance task (Figure 4A). We compared the effects of double-knockout (ι/ζ -dKO) to single knockout (ζ -KO), by injecting $PKC\iota^{fl/fl}$ -PKM $\zeta^{-/-}$ littermates bilaterally in hippocampus with either AAV-Cre or AAV-eGFP. Mice received 3, 30-min training trials separated by 24 hours and a final retention test without shock the next day. Our measure of short-term memory was maximum avoidance time within each session (Figure 4B). Maximum avoidance time reflects the time between shocks, which is controlled by the animal's behavior and within-trial memory. Note that compared to pretraining inter-shock zone entries, the maximum avoidance time for ζ -KOs and ι/ζ -dKOs greatly increased. This indicates that they both had short-term memory for the shock zone. Our measure of long-term memory was time to first entry into the shock zone at the beginning of each session (Figure 4C). This reflects their long-term memory maintained across days from previous trials. ζ -KOs first entry times increased dramatically from pretraining to both session 3 and retention test, indicating that mice with PKC ι maintain long-term memory. In contrast, ι/ζ -dKO mice displayed a minimal increase at session 3 that was not significantly different from pretraining, and no detectable difference between pretraining and the retention test, indicating minimal or no spatial long-term memory.

Discussion

Although the structural model of memory storage is the standard account (Bailey and Kandel, 1993), the maintenance of hippocampal LTP and long-term memory in WT mice appeared to crucially depend on a persistent biochemical mechanism involving an aPKC, PKM ζ (Pastalkova et al., 2006; Shema et al., 2007; Wang et al., 2016; Gao et al., 2018; Tsokas et al., 2024). However, both long-term phenomena are observed when PKM ζ is genetically deleted (Lee et al., 2013; Volk et al., 2013; Tsokas et al., 2016; Tsokas et al., 2024). These results either confirm the structural model or point to the existence of a compensatory biochemical mechanism for memory (Tsokas et al., 2016).

Here we found that a second biochemical mechanism for memory comes into play to compensate for the loss of the first. The PKC isoform most closely related to PKM ζ , PKC ι , which normally plays only a transient role in LTP and short-term memory, becomes persistently active in LTP and long-term memory when PKM ζ is genetically ablated (Figures 1, 2). When we eliminate the two persistent biochemical processes by double-knockout there is no enduring hippocampal LTP or long-term memory (Figures 3, 4). These results establish a biochemical mechanism as the leading candidate for how a memory endures, and identifies both PKM ζ and PKC ι as persistent kinases crucial for sustaining memory, beyond the contribution of any static structural processes necessary for synaptic integrity.

The ι/ζ -dKO exhibits an early transient LTP that could not be sustained into late-phase persistent LTP (Figure 3). In addition, bilateral hippocampal ι/ζ -dKO eliminated long-term spatial memory but did not prevent learning, short-term memory, or expression of the place avoidance behavior (Figure 4). In WT mice, PKC ι is a key contributor to early-LTP and short-term memory (Ren et al., 2013; Wang et al., 2016); therefore, there must be additional compensation for short-term processes present in the aPKC-dKO.

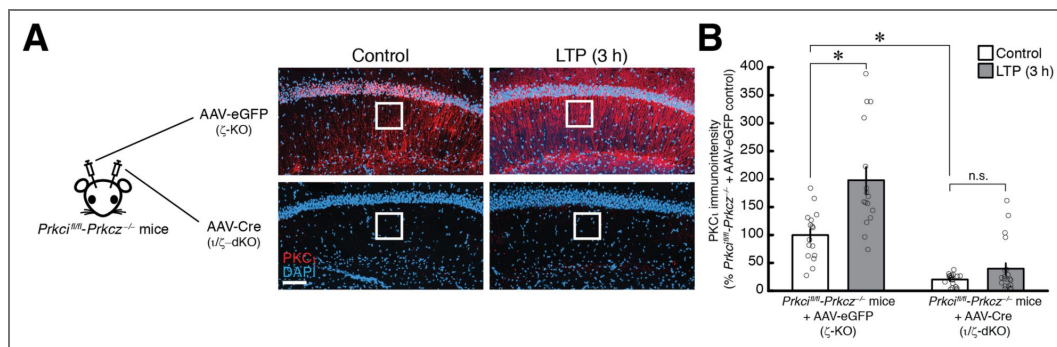


Figure 2. Compensatory increases of PKCι during hippocampal late-LTP maintenance in *Prkci^{fl/fl}-Prkcz^{-/-}* mice.

(A) Left, schematic of experimental protocol shows AAV expressing Cre by CMV promoter injected into ipsilateral hippocampus of a *Prkci^{fl/fl}-Prkcz^{-/-}* mouse, and control AAV expressing eGFP in contralateral hippocampus. Hippocampal slices are prepared 3 weeks later. Right, representative images of PKCι-immunohistochemistry from adjacent slices in AAV-eGFP-injected (ζ-KO) hippocampus show PKCι persistently increases 3 h post-tetanzation (top row), and low, unchanging levels of PKCι in the AAV-Cre-injected (ι/ζ-dKO) hippocampus (bottom row). White boxes show *st. radiatum* regions of interest. (B) Mean ± SEM. The two-way ANOVA reveals the main effects of treatment (AAV-Cre [ι/ζ-dKO] vs. AAV-eGFP [ζ-KO], $F_{1,68} = 83.58$, $P < 0.00001$, $\eta^2_p = 0.55$), and stimulation (HFS vs. test, $F_{1,68} = 20.47$, $P = 0.00003$, $\eta^2_p = 0.23$), and an interaction of treatment x stimulation ($F_{1,68} = 9.09$, $P = 0.004$, $\eta^2_p = 0.12$). *Post-hoc* analysis confirms that, compared to the ζ-KO control group, the intensity of PKCι immunoreactivity was significantly decreased in ι/ζ-dKO (P 's < 0.002 for both control and LTP in ι/ζ-dKO), and increased in ζ-KO after HFS ($P = 0.00011$, ζ-KO, n 's = 16, ι/ζ-dKO, n 's = 20). Intensity of PKCι immunoreactivity did not change in the ι/ζ-dKO between the control and HFS groups ($P = 0.3$). Bar = 100 μm.

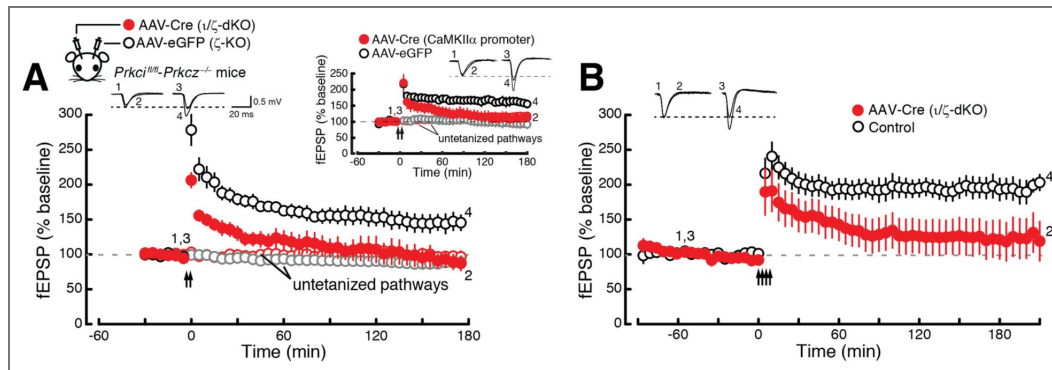


Figure 3. ι/ζ -dKO hippocampus shows transient LTP, but not persistent LTP.

(A) Late-LTP is absent in ι/ζ -dKO hippocampus. Above left inset, schematic of intrahippocampal injections of AAC-Cre and AAV-eGFP. Middle inset, representative fEPSPs correspond to numbered times in time-course below. Below, filled red circles, AAC-Cre expressed by CMV promoter and HFS with 2 tetanic trains; open red circles, test stimulation of a second synaptic pathway within the hippocampal slice. HFS tetani shown at arrows. Open black circles, AAV expressing eGFP with HFS; open grey circles, with test stimulation. Three-way mixed-design ANOVA reveals main effects of treatment (hippocampal injections of AAV-Cre [ι/ζ -dKO] vs. AAV-eGFP [ζ -KO], $F_{1,20} = 8.45$, $P = 0.0009$, $\eta^2_p = 0.30$), and stimulation (HFS vs. test stimulation, $F_{1,20} = 5.90$, $P = 0.025$, $\eta^2_p = 0.23$), as well as a 3-way interaction among treatment $\times$ stimulation $\times$ time (5-min average of pre-HFS and 3-h post-HFS, $F_{1,20} = 12.68$, $P = 0.002$, $\eta^2_p = 0.39$). *Post-hoc* analysis confirms established LTP is not maintained in ι/ζ -dKO 3 h after HFS when compared to pre-HFS basal responses ($P = 0.7$). *Post-hoc* analysis also confirms the control hippocampus maintains established LTP ($P = 0.0002$). Test stimulation was unaffected by AAV-Cre or AAV-eGFP injections ($P = 0.9$ and $P = 0.7$, respectively, n 's = 6). Right inset, ι/ζ -dKO by CaMKII α promoter expression of Cre recombinase eliminates late-LTP. Three-way mixed-designed ANOVA reveals interaction between treatment (ζ -KO vs. ι/ζ -dKO) and stimulation (HFS vs. test stimulation, $F_{1,14} = 6.62$, $P = 0.02$, $\eta^2_p = 0.32$), and a 3-way interaction among treatment, stimulation, and time (5 min pre-HFS and 3 h post-HFS, $F_{1,14} = 8.56$, $P = 0.01$, $\eta^2_p = 0.38$). *Post-hoc* analysis confirms that compared to pre-HFS basal responses, LTP is not maintained in ι/ζ -dKO hippocampus 3 h post-HFS ($P = 0.8$) and is maintained in the control hippocampus ($P = 0.003$). Test stimulation was unaffected by AAV-Cre or AAV-eGFP injections ($P = 0.4$ and $P = 0.9$, respectively). ι/ζ -dKO HFS, $n = 5$; ι/ζ -dKO test, $n = 4$; ζ -KO HFS, $n = 5$; ζ -KO test, $n = 4$. **(B)** LTP does not persist in ι/ζ -dKO mice after the stronger tetanization. ANOVA with repeated measurements reveals main effects of time (5 min pre-HFS, 20 min post-HFS, and 3 h post-HFS, $F_{2,14} = 20.51$, $P < 0.0001$, $\eta^2_p = 0.75$). *Post-hoc* analysis confirms that early-LTP is established in both ι/ζ -dKO and control groups (5 min pre-HFS vs. 20 min post-HFS, $P = 0.005$ and 0.002 , respectively), and no difference between these two groups at 20 min post-HFS ($P = 0.6$). However, LTP in ι/ζ -dKO did not persist 3 h (5 min pre-HFS vs. 3 h post-HFS, $P = 0.4$), whereas LTP is intact in control. ($P = 0.008$). The ι/ζ -dKO, $n = 5$; control, $n = 4$.

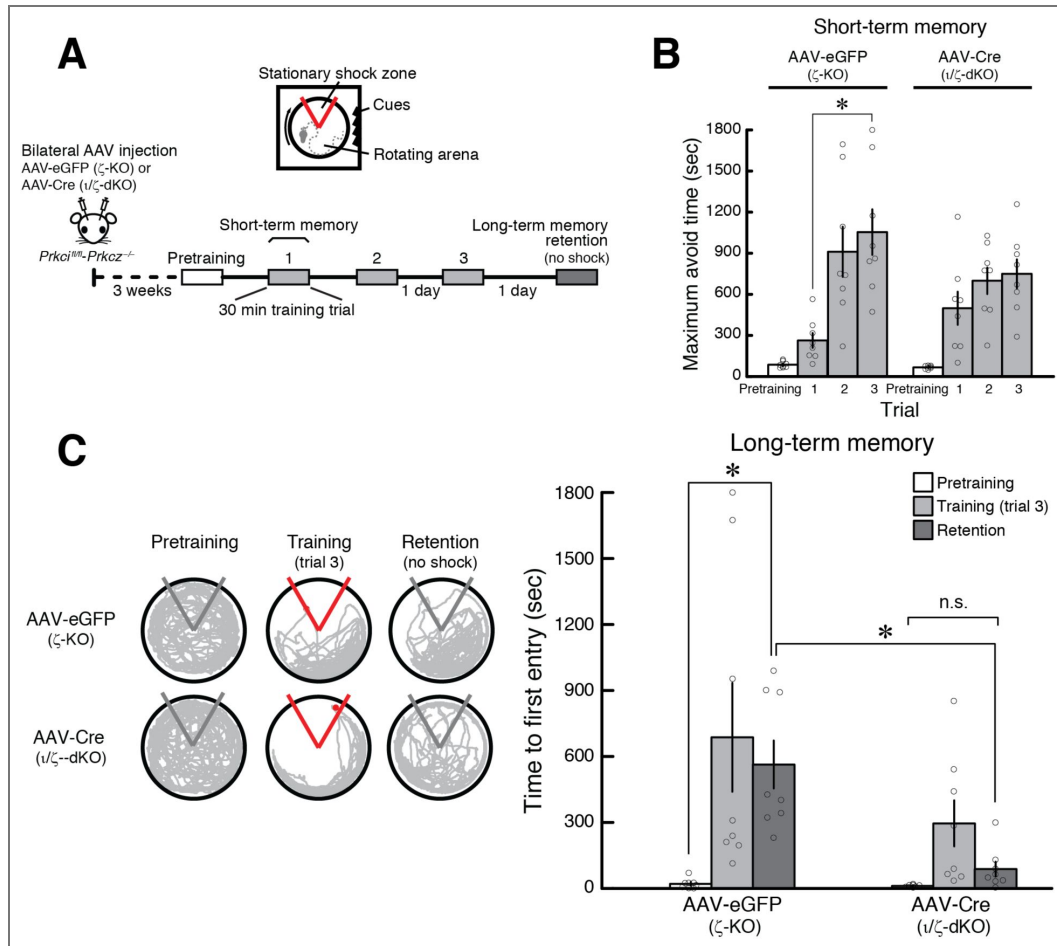


Figure 4. Impaired spatial long-term memory and intact spatial short-term memory in mice with bilateral hippocampal/ζ-dKO.

(A) Above, schematic of active place avoidance training apparatus shows a slowly rotating arena containing a nonrotating shock zone sector (delineated in red). Visual cues located on the walls of the room are needed to avoid the shock zone. Below, protocol for active place avoidance. *Prkci^{fl/fl}-Prkcz^{-/-}* mice are injected bilaterally in hippocampus with AAV-Cre (u/ζ-dKO) or AAV-eGFP (ζ-KO, control). (B) u/ζ-dKO does not affect short-term memory as assessed by maximum avoidance time during the first training trial. The contrast analysis reveals that the increases of maximum avoidance time from pretraining to trial 1 are not different between AAV-eGFP-injected and AAV-Cre-injected groups ($t_{14} = 1.91$, $P = 0.08$, $d = 1.91$), indicating both groups of mice successfully established short-term memory. However, the improvement of maximum avoidance time from trial 1 to trial 3 are different between the groups ($t_{14} = 2.93$, $P = 0.01$, $d = 2.88$), suggesting the two groups performed differently between daily training sessions. In addition, the ANOVA with repeated measurement discovers no group effect (AAV-eGFP-injected vs. AAV-Cre-injected, $F_{1,14} = 0.56$, $P = 0.47$, $\eta^2_p = 0.04$), but significant effects of trial ($F_{3,42} = 30.37$, $P < 0.0001$, $\eta^2_p = 0.68$), and interaction ($F_{3,42} = 2.93$, $P = 0.04$, $\eta^2_p = 0.17$). *Post-hoc* tests confirm that the maximum avoidance time in trial 1 is not different between the two groups ($P = 0.14$). The AAV-eGFP-injected group improved their performance between trial 1 and trial 3 ($P = 0.0002$), whereas the AAV-Cre showed no improvement ($P = 0.2$; n 's = 8). These data indicate no differences in short-term memory between AAV-eGFP- and AAV-Cre-injected groups, but only the AAV-Cre-injected group failed to improve between daily trials, suggesting inability to retain avoidance memory across days. (C) PKC ι gene ablation impairs long-term memory in *Prkci^{fl/fl}-Prkcz^{-/-}* mice. Left, representative paths during 10-min of pretraining, at end of training trial 3, and 1-day memory retention. Right, mean $\pm$ SEM. The ANOVA with repeated measurement finds main effects of group (AAV-eGFP vs. AAV-Cre, $F_{1,14} = 10.53$, $P = 0.006$, $\eta^2_p = 0.43$) and training phase (pretraining, trial 3 of training, retention, $F_{2,28} = 7.65$, $P = 0.002$, $\eta^2_p = 0.35$). *Post-hoc* analysis reveals that the mice with AAV-Cre-injected u/ζ-dKO hippocampus perform poorer during the memory retention test, compared to AAV-eGFP-injected littermates ($P = 0.02$). The mice with u/ζ-dKO hippocampus show no difference between the memory retention test and pretraining trial ($P = 0.9$), whereas the AAV-eGFP-injected mice show long-term memory is maintained ($P = 0.02$; n 's = 8). In addition, pretraining vs. training trial 3 was significantly different in ζ-KO ($P = 0.006$), but not in u/ζ-dKO ($P = 0.4$).

Our finding that PKC ι compensates for the absence of PKM ζ raises the question, how is it induced. Compensatory gene expression can be induced after gene knockout by the fragments of mRNA that are produced by transcription upstream of the site of Cre-recombination (El-Brolosy and Stainier, 2017 [\[3\]](#); El-Brolosy et al., 2019 [\[4\]](#); Ma et al., 2019 [\[5\]](#)). This could explain why PKM ζ -KOs produce compensation, but PKM ζ -shRNA or PKM ζ -antisense oligodeoxynucleotides that retain full-length PKM ζ mRNA transcription do not (Tsokas et al., 2016 [\[6\]](#); Wang et al., 2016 [\[7\]](#)). In medial prefrontal cortex, both early- and late-LTP are absent in PKM ζ -KO mice (Kniffin et al., 2025 [\[8\]](#)). This suggests that the PKC ι activation mediating early-LTP in the WT hippocampus may not be available as a biochemical process to compensate for loss of PKM ζ in the ζ -KO medial prefrontal cortex (Sacktor, 2025 [\[9\]](#)).

Once increased, how does PKC ι accomplish maintenance? PKM ζ maintenance is linked to the kinase's second messenger-independent, persistent activity (Sacktor et al., 1993 [\[10\]](#)). PKM ζ is autonomously active because the kinase is an independent catalytic domain that lacks the PKC ζ autoinhibitory regulatory domain (Sacktor et al., 1993 [\[10\]](#); Hernandez et al., 2003 [\[11\]](#)). PKC ι , however, is a full-length PKC isoform with a regulatory domain that inhibits its catalytic domain. Therefore, for PKC ι to compensate for PKM ζ the aPKC likely requires additional posttranslational mechanisms to persistently activate and localize the kinase at active synapses (Figure 1C [\[12\]](#)). Unlike other isoforms, PKC ι can be activated by its regulatory domain binding to postsynaptic proteins such as p62 (Jiang et al., 2009 [\[13\]](#); Ren et al., 2013 [\[14\]](#)). This activation of PKC ι by protein-protein interaction may sustain its kinase activity longer than the transient lipid second messengers that activate the conventional/novel PKCs, allowing one persistent biochemical mechanism to replace the other.

The maintenance properties of PKM ζ not only depend on continuous activity but also on continuous binding to the postsynaptic scaffolding protein KIBRA/WWC1 (kidney and brain protein/WW and C2 domain protein 1) (Tsokas et al., 2024 [\[15\]](#)). This interaction localizes PKM ζ to active synapses in a process called persistent synaptic tagging (Tsokas et al., 2024 [\[15\]](#); Shouval et al., 2025 [\[16\]](#)). KIBRA also weakly binds PKC ι (Tsokas et al., 2024 [\[15\]](#)). In WT mice, the stronger binding of PKM ζ to KIBRA might allow it to replace PKC ι at active synapses in the transition from early- to late-LTP. In the hippocampus of PKM ζ -KO mice, PKC ι at active synapses would not be replaced. PKM ζ and PKC ι also compete for binding to PAR3 (partitioning defective protein 3), another postsynaptic protein that localizes aPKCs within neurons (Parker et al., 2013 [\[17\]](#); Zhang and Wei, 2022 [\[18\]](#)). Thus, when PKM ζ is genetically deleted, synaptic tags such as KIBRA and PAR3 that are components of the persistent PKM ζ mechanism may now shift to PKC ι to maintain hippocampus-dependent LTP and long-term memory.

Materials and Methods

Reagents

Unless otherwise stated, reagents were from MilliporeSigma.

Animals

Male mice were on C57BL/6 background and at least 4-months-old for all experiments. The PKM ζ -null mouse line was previously described (Lee et al., 2013 [\[19\]](#)) and provided by Robert O. Messing (Univ. Texas at Austin, TX, USA). Conditional PKM ζ and PKC ι mice were generated by Sourav Ghosh as previously described (Scott et al., 2019 [\[20\]](#); El Allam et al., 2024 [\[21\]](#); Mercu et al., 2024 [\[22\]](#)). *Camk2a-CreER^{T2}* mice were from Jackson Labs. Tamoxifen vehicle for i.p. injections was sunflower seed oil.

Hippocampal slice recording and stimulation

Acute mouse hippocampal slices (450 μ m) were prepared as previously described (Tsokas et al., 2016 [\[6\]](#); Tsokas et al., 2019 [\[23\]](#)). Hippocampi were dissected, bathed in ice-cold dissection buffer, and sliced with a McIlwain tissue slicer in a cold room (4°C). The dissection buffer contained (in mM): 125 NaCl, 2.5 KCl, 1.25 NaH₂PO₄, 26 NaHCO₃, 11 glucose, 10 MgCl₂, and 0.5 CaCl₂, and was

bubbled with 95% O₂/5% CO₂ to maintain pH at 7.4. After dissection the slices were transferred to an Oslo-type interface recording chamber (31.5 ± 1°C) (Tsokas et al., 2019). The recording superfusate consisted of (in mM): 118 NaCl, 3.5 KCl, 2.5 CaCl₂, 1.3 MgSO₄, 1.25 NaH₂PO₄, 24 NaHCO₃, and 15 glucose, bubbled with 95% O₂/5% CO₂, with a flow rate of 0.5 ml/min.

Field EPSPs were recorded with a glass extracellular recording electrode (2–5 MΩ) placed in the CA1 *st. radiatum*, and concentric bipolar stimulating electrodes (CBBRE75 and 30200; FHC, Bowdoin, ME) were placed on either side within CA3 or CA1. Test stimulation rate was once every 30 sec, alternating every 15 sec between stimulating electrodes. Based upon a pre-established exclusion criterion, a slice was not used if fEPSP spike threshold was <2 mV on initial input-output analysis. Pathway independence was confirmed by the absence of paired-pulse facilitation between the two pathways. A single stimulating electrode was used for immunocytochemistry with a test stimulation rate of once every 30 sec. HFS optimized to produce a relatively rapid onset of protein synthesis-dependent late-LTP, consisted of two 100 Hz-1 s tetanic trains, at 25% of spike threshold, spaced 20 sec apart (Tsokas et al., 2005). HFS optimized to produce maximal late-LTP consisted of four 100 Hz-1 s tetanic trains, at 25% of spike threshold, spaced 5 min apart (Scharf et al., 2002; Serrano et al., 2005). The maximum slope of the rise of the fEPSP was analyzed on a PC using the WinLTP data acquisition program (Anderson and Collingridge, 2007).

Immunoblots and Immunohistochemistry

Immunoblots of total hippocampus were performed as previously described, using antibodies in Key Resources Table (Tsokas et al., 2016). Immunoblots were stained with multiple antisera to visualize multiple PKCs on the same immunoblot. To conserve antisera the immunoblots were cut to isolate the relevant proteins based on molecular weight. Isoforms with similar molecular weights were either stained with antisera of different species or on separate blots.

Quantitative immunohistochemistry for Figure 2 and Figure 3 — figure supplement 1 was as described (Hsieh et al., 2021; Tsokas et al., 2024), using mouse anti-PKC ϵ primary antibody (1:1000, E-7, Santa Cruz SC-376344).

Immunohistochemistry for Figure 1C was performed as follows. Free-floating sections were permeabilized with phosphate-buffered saline (PBS) containing 0.1% Tween20 (PBS-T) for 1 h at room temperature and blocked with 10% normal goat serum in PBS-T (blocking buffer) for 2.5 h at room temperature. One batch of sections was incubated overnight at 4 °C with rabbit anti-PKM ζ C-2 antisera primary antibody (1:1,000) (Hernandez et al., 2003) and a second batch of sections with rabbit anti-PKC ϵ (1:1000, Cell Signaling #2998S) in blocking buffer. After washing 3 times for 10 min each in PBS-T, both batches of sections were incubated with the secondary antibody goat anti-rabbit-Alexa 647 (1:500 in blocking buffer; Jackson ImmunoResearch) for 2 h at room temperature. After washing 3 times for 10 min each in PBS-T and extensive washing with PBS, the sections were mounted with Vectashield with 4',6-diamidino-2-phenylindole (DAPI, Vector Laboratories). Three sections from each mouse were examined using an upright Leica SP8 confocal microscope and analyzed using ImageJ (version 1.53a). For each section, 8.5 μ m-thick Z-stacks of the dorsal CA1 were created using the maximum intensity projection function in ImageJ. For each *st. pyramidale*, *radiatum*, and *lacunosum-moleculare*, two square regions of interest were centered in each stack. Measurements were made from each mouse in each region of interest. The raw integrated density (defined as the sum of the values for all pixels) of the Z-stack region of interest expressing the fluorescent label was measured for the volume of target pixels, and the average of each measurement was taken as representative for the region and each mouse.

AAV injections

Mice were anesthetized in a closed chamber filled with the inhalation anesthetic isoflurane (RWD Life Science, R510-22-10), and then fixed in a stereotaxic apparatus (Stoelting Co.). Anesthesia was maintained with isoflurane inhalation (1%–2.5% via trachea). The eyes of the mice were safeguarded using erythromycin ophthalmic ointment (0.5%). The skull was exposed and cleaned using 3% hydrogen peroxide. Small holes in the skull were then drilled with the following stereotaxic coordinates: left hippocampus (triple injection: AP: -1/ ML: -0.7/ DV: - 1.65; AP: -1.8/ ML:

-1.5/ DV: -2; AP: -2.7/ ML: -2/ DV: -2, below the skull surface) and right hippocampus (triple injection: AP: -1/ ML: +0.7/ DV: -1.65; AP: -1.8/ ML: +1.5/ DV -2; AP: -2.7/ ML: +2/ DV: -2, below the skull surface). The virus was injected using a 34-gauge needle with a Hamilton syringe at 0.1 μ l/min rate into target regions. At all injected points, the tip of the needle was positioned 0.05 mm below the target coordinate and returned to the target site after 2 min. After injection, the needle stayed in place for an additional 7 min and was slowly withdrawn. AAVs expressing Cre-recombinase and eGFP were from Addgene. For physiology, 0.5 μ l of virus pENN.AAV.CMV.PI.Cre.rBG (AAV2.9) (1×10^{13} viral genomes [vg]/ml) was injected into one CA1, and 0.5 μ l of virus pAAV.CMV.PI.EGFP.WPRE.bGH (AAV2.9) (1×10^{13} vg/ml) was injected in the contralateral side. Virus pENN.AAV.CamKIIa 0.4 Cre SV40 (AAV2.9) (1×10^{13} vg/ml) was used to express Cre-recombinase by the CaMKIIa promoter.

Conditioning

All experiments were performed blindly. Active place avoidance was conducted with a commercial computer-controlled system (Bio-Signal Group, Acton, MA). The mouse was placed on a 40-cm diameter circular arena rotating at 1 rpm. The specialized software, Tracker (Bio-Signal Group, Acton, MA), was used to detect the animal's position 30 times per second by video tracking from an overhead camera. A clear wall made from polyethylene terephthalate glycol-modified (PET-G) was placed on the arena to prevent the animal from jumping off the elevated arena surface. A 5-pole shock grid was placed on the rotating arena, and the shock was scrambled across the 5-poles when the mouse entered the shock zone. All experiments used the "Room+Arena-" task variant that challenges the mouse on the rotating arena to avoid a shock zone that was a stationary 60° sector (Pastalkova et al., 2006 [DOI](#)). Every 33 ms, the software determined the mouse's position, whether it was in the shock zone, and whether to deliver shock. After the animal enters the shock zone for 500 ms, a constant current foot-shock (60 Hz, 500 ms) was delivered and repeated with the interval of 1500 ms until the mouse left the shock zone. The shock intensity was 0.2 or 0.3 mA, which was the minimum amplitude to elicit flinch or escape responses. The animal was forced to actively avoid the designated shock zone because the arena rotation periodically transported it into the shock area.

The tracked animal positions with timestamps were analyzed offline (TrackAnalysis, Bio-Signal Group, Acton, MA) to extract several end-point measures. The time to first enter the shock zone estimates ability to avoid shock and was taken as an index of between-session long-term place avoidance memory. The maximum time without receiving a shock within each session estimates the within-session short-term place avoidance memory. A pretraining period on the apparatus equivalent in time to a training session, but without shock, was provided, and the time to first entry and between pretraining entrances into the shock zone with the shock off was recorded.

The training schedule was as follows: 1 day after a 30-min pretraining session, the animals received three 30-min training trials, with an intertrial interval of 1 day. Long-term memory retention was tested the following day without shock. Pre-established exclusion criterion was if cannulae were found to be incorrectly targeted. No mice were excluded.

Statistics

Sample sizes vary for the different experimental approaches (biochemistry, extracellular field potential physiology, and behavior). The hypothesis that PKM ζ is compensated predicts all-or-none effects in the experiments, and this provided a basis for sample size estimates. Power analyses were performed using G*Power Version 3.1.9.7 with $\alpha = 0.05$ and $\beta = 0.8$ and large effect sizes of 1.5–2.0. The effect size estimates were based on prior studies that demonstrated essentially all-or-none effects of PKM ζ inhibition on the immunoblot, immunohistochemical, physiological, and behavioral assays used here (Tsokas et al., 2016 [DOI](#); Hsieh et al., 2021 [DOI](#); Tsokas et al., 2024 [DOI](#)). Two-population Student *t* tests with Bonferroni corrections were performed to compare protein levels by immunoblot and immunocytochemistry in the PKM ζ -cKO and control mice. For LTP experiments the responses to test stimuli were averaged across 5 min for statistical comparisons. Repeated measures ANOVA was used to compare the change in the potentiated response at the

time points described. Multi-factor comparisons were performed using mixed-design ANOVA with repeated measures, as appropriate. The degrees of freedom for the critical t values of the t tests and the F values of the ANOVAs are reported as subscripts. *Post-hoc* multiple comparisons were performed by Newman-Keuls tests as appropriate. Statistical significance was accepted at $P < 0.05$. Effect sizes for binary comparisons and one-way ANOVAs are reported as Cohen's d and as η^2_p for multi-factor ANOVA effects.

Supplementary Materials

Antibodies for immunoblotting	Vendor and catalog number	Concentration
aPKC		
PKM ζ (C2)	Sacktor Lab (<i>Hernandez et al., 2003</i>)	1:20,000
PKC ι (E-7)	Santa Cruz sc-376344	1:200
p-PKC		
p-T410 PKC ζ (H2)	Santa Cruz sc-271962	1:200
Phospho-PKC (pan) (ζ Thr410) (190D10)	Cell Signaling Technology #2060	1:100
cPKC		
α (H-7)	Santa Cruz sc-8393	1:200
β I (E-3)	Santa Cruz sc-8049	1:500
β II (F-7)	Santa Cruz sc-13149	1:100
γ (C-4)	Santa Cruz sc-166385	1:200
n-PKC		
δ [EPR17075]	Abcam ab182126	1:100
ϵ (E-5)	Santa Cruz sc-1681	1:1000
η [EPR18513]	Abcam ab179524	1:200
θ (E-7)	Santa Cruz sc-1680	1:100
Other		
CaMKII α	ThermoFisher (Invitrogen) 137300	1:100
p-CaMKII [T286]	Cell Signaling Technology #3361	1:100
Actin	Sigma A4700	1:5000

[Key resources table.](#)

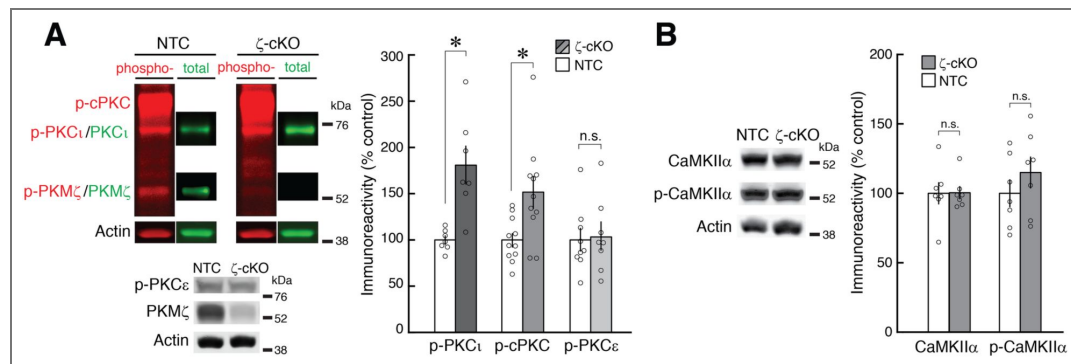


Figure 1 — figure supplement 1. ζ -cKO increases activation-loop-phosphorylation state of atypical PKC ι and conventional PKCs, but not novel PKC ϵ or CaMKII α autophosphorylation. (A) Left, above, representative immunoblots show increases in activation-loop phosphorylated PKC ι (p-PKC ι) and conventional-PKCs (p-cPKCs) in PKM ζ -cKO mice. Red, phospho-PKCs; green, total PKCs from the same samples. Mr's shown at right. Below, p-PKC ϵ , recognized by its higher Mr, does not change. Right, mean $\pm$ SEM **(B)** Levels of total CaMKII α and T286-autophosphorylated CaMKII α do not change in ζ -cKO hippocampus. Statistics in Figure 1 — table supplement 2.

Figure 1 — figure supplement 2. Compensation for spatial memory in ζ -cKO mice.

Two-way ANOVA (treatment X training) revealed a significant effect of training ($F_{1,662, 9,972} = 41.93, P < 0.0001$), but not an effect of treatment ($F_{1,6} < 0.001, P = 1.0$) or their interaction ($F_{2, 12} = 0.48, P = 0.6$). Further comparisons using Bonferroni-corrected tests revealed significant differences between pretraining and training (Trial 3) ($P = 0.002$) and pretraining and retention ($P = 0.001$), but no differences between training (Trial 3) and retention ($P = 0.1$). Moreover, comparison also revealed significant differences between pretraining and training (Trial 3) in both vehicle and 4-OH tamoxifen groups ($P = 0.02$ and $P = 0.01$, respectively) and between pretraining and retention in both vehicle and 4-OH tamoxifen groups ($P = 0.03$ and $P = 0.05$, respectively), confirming the treatment groups did not behave differently.

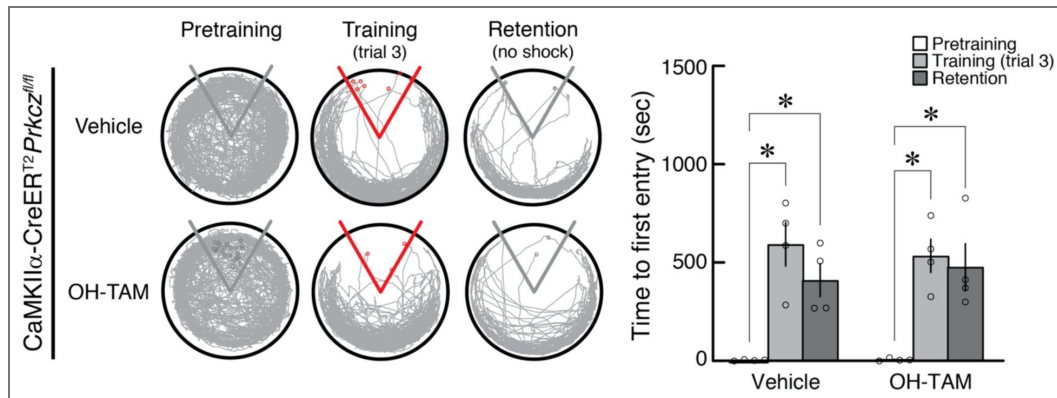


Figure 3 — figure supplement 1.

(A) Representative immunohistochemistry of AAV expressing Cre-recombinase by CaMKII α promoter in $Prkci^{fl/fl}$ - $Prkcz^{-/-}$ mice shows loss of PKC ζ in ζ -cKO hippocampus and compensatory increase in PKC ι during LTP maintenance in contralateral control eGFP-injected hippocampus. Left, schematic of sites of injection; right, PKC ζ immunohistochemistry. DAPI stains nuclei in CA1. Bar = 100 μ m. **(B)** ζ -cKO hippocampus shows compensated LTP. AAV expressing Cre-recombinase by CaMKII α promoter was injected into $Prkci^{fl/fl}$ - $Prkcz^{+/+}$ mice. Loss of early-LTP is compensated in the PKC ζ -cKO as in previous reports (Sheng et al., 2017). Paired t-test reveals early-LTP is established at 20 min post-tetanzation ($t_3 = 4.76, P = 0.02$, Cohen's $d = 3.41, n = 4$).

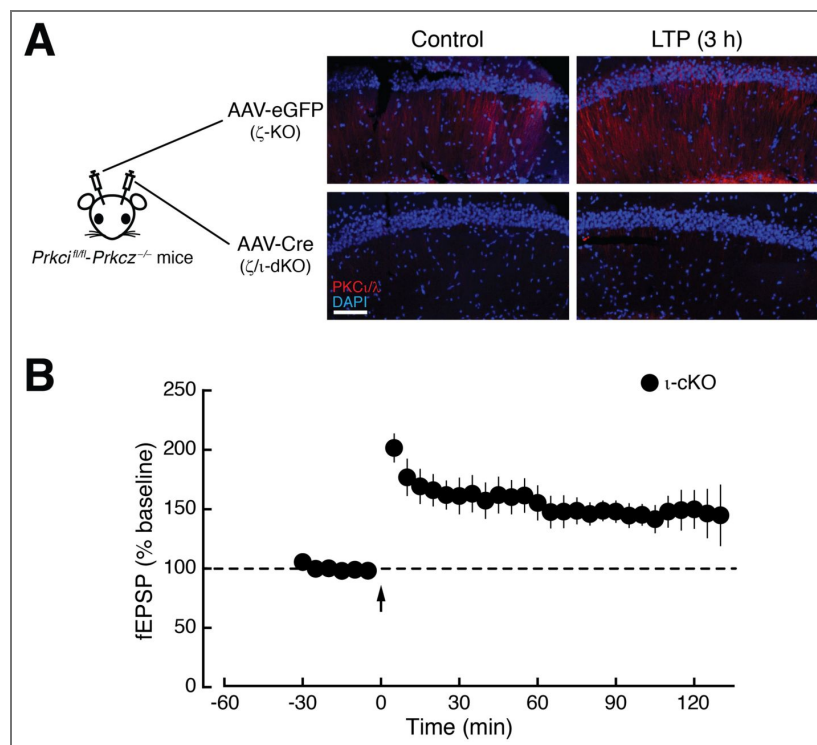


Table S1. Statistics for data presented in Figure 1

Significant differences with Bonferroni correction are in bold.

Isozyme / region	Non-transgenic control (NTC)		ζ -cKO		Degree of freedom	<i>t</i>	<i>P</i>	Cohen's <i>d</i>
	Mean $\pm$ SEM	n	Mean $\pm$ SEM	n				
(A)								
aPKC								
PKM ζ	100 $\pm$ 4.9	9	11.8 $\pm$ 2.4	8	15	15.4	< 0.0001	7.5
PKC ι	100 $\pm$ 7.3	4	172.4 $\pm$ 18.4	5	7	3.9	0.01	2.2
(B)								
cPKC								
α	100 $\pm$ 2.3	8	140.0 $\pm$ 7.0	8	14	5.4	< 0.0001	2.7
β I	100 $\pm$ 6.2	8	142.1 $\pm$ 10.6	8	14	3.4	0.004	1.7
β II	100 $\pm$ 6.3	6	150.2 $\pm$ 5.5	5	9	5.9	0.0002	3.6
γ	100 $\pm$ 8.6	9	149.6 $\pm$ 15.4	9	16	2.8	0.01	1.3
nPKC								
δ	100 $\pm$ 10.4	7	108.2 $\pm$ 10.3	7	12	0.6	0.6	0.3
ϵ	100 $\pm$ 5.3	9	114.8 $\pm$ 11.0	9	16	1.2	0.2	0.6
η	100 $\pm$ 17.1	6	109.0 $\pm$ 10.9	6	10	0.4	0.7	0.3
θ	100 $\pm$ 10.1	6	97.7 $\pm$ 14.9	7	11	0.1	0.9	0.1
(C)								
PKM ζ								
<i>pyramidale</i>	100 $\pm$ 13.6	5	14.1 $\pm$ 2.1	5	8	6.2	0.0003	3.9
<i>radiatum</i>	100 $\pm$ 3.6	5	23.6 $\pm$ 3.9	5	8	14.4	<0.0001	9.1
<i>lac-mol</i>	100 $\pm$ 22.0	5	54.2 $\pm$ 13.2	5	8	1.8	0.1	1.1
PKC ι								
<i>pyramidale</i>	100 $\pm$ 9.8	5	235.3 $\pm$ 26.4	5	8	4.8	0.001	3.0
<i>radiatum</i>	100 $\pm$ 36.9	5	411.2 $\pm$ 26.2	5	8	6.9	0.0001	4.3
<i>lac-mol</i>	100 $\pm$ 40.1	5	226.7 $\pm$ 26.7	5	8	2.6	0.03	1.7

Table S2. Statistics for data presented in Figure 1 — figure supplement 1

Significant differences with Bonferroni correction are in bold.

Kinase	Mean $\pm$ S.E.M.	<i>n</i>	Mean $\pm$ S.E.M.	<i>n</i>	Degree of freedom	<i>t</i>	<i>P</i>	Cohen's <i>d</i>
(A)								
p-PKC								
p-PKC ι	100 $\pm$ 4.0	7	181.1 $\pm$ 19.8	7	12	4.0	0.002	2.1
p-cPKC	100 $\pm$ 7.1	11	151.4 $\pm$ 16.4	11	20	3.3	0.009	1.2
p-PKC ϵ	100 $\pm$ 11.7	9	104.0 $\pm$ 14.2	8	15	0.2	0.8	0.1
(B)								
CaMKII α	100 $\pm$ 7.63	7	96.33 $\pm$ 1.95	6	11	0.43	0.67	0.24
p-CaMKII α	100 $\pm$ 9.65	7	114.94 $\pm$ 11.17	7	12	1.01	0.33	0.54

Data availability

All data are available in the main text or the supplementary materials.

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Author contributions

Conceptualization: TCS, AAF

Methodology: PT, CH, AG-P, LK, SG

Investigation: PT, CH, AG-P, LK, LMR-V, DAC, KDA, HJHS, SK, BJW, SS, REF-O

Visualization: CH, AG-P

Funding acquisition: TCS, AAF, JEC

Project administration: TCS, AAF

Supervision: TCS, AAF

Writing – original draft: TCS, AAF, PJB, JR

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The Garry and Sarah S. Sklar Fund		Panayiotis Tsokas

Author ORCID iDs

Panayiotis Tsokas: <https://orcid.org/0000-0002-1679-8980>

Changchi Hsieh: <https://orcid.org/0000-0003-0488-1565>

Alejandro Grau-Perales: <https://orcid.org/0000-0002-0850-4313>

Kim D Allen: <https://orcid.org/0000-0001-9774-5014>

Rafael E Flores-Obando: <https://orcid.org/0000-0002-0736-9527>
Peter John Bergold: <https://orcid.org/0000-0002-6335-1380>
Jerry Rudy: <https://orcid.org/0000-0002-8212-5602>
André Antonio Fenton: <https://orcid.org/0000-0002-5063-1156>
Todd Charlton Sacktor: <https://orcid.org/0000-0002-8625-8701>

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Peer reviews

Reviewer #1 (Public review):

Summary:

An ongoing controversy in the field of learning and memory is the specific neural mechanism that maintains long-term memory (LTM). A prominent hypothesis proposed by Sacktor and Fenton and their colleagues is that LTM is maintained by the ongoing activity of the atypical PKC isoform PKM ζ . Early evidence in support of this hypothesis came from experiments showing that an inhibitory peptide, ZIP, whose activity was purported to be specific for PKM ζ , blocked late-phase hippocampal LTP (L-LTP) and LTM. However, in 2013, two articles reported that LTM was normal in PKM ζ knockout mice and that ZIP erased LTM in the knockout mice, indicating that ZIP lacked specificity for PKM ζ . In response, Sacktor and Fenton and colleagues reported in 2016 that in PKM ζ null mice, there is an increase in the expression of PKC δ /λ, a related isoform of atypical PKC, and this increased expression can compensate for PKM ζ ; their data indicated that the upregulation of PKC δ /λ mediates L-LTP and LTM in the PKM ζ . In the present article, the authors provide additional support for this idea. They replicate the finding of an upregulation of PKC δ /λ expression in the hippocampus of PKM ζ knockout mice; in addition, they show that the expression of several other PKC isoforms is upregulated in the knockouts. They find that down-regulation of PKC δ /λ expression in the hippocampus using the Cre-LoxP technology, the 2016 paper merely used an inhibitor to block the activity of PKC δ /λ-blocks L-LTP. Finally, the authors demonstrate that, although LTM is preserved in the single PKM ζ knockout mouse, it is eliminated in the PKM ζ /PKC δ /λ double knockout mouse.

Strengths:

The experiments appear to have been carefully executed, the results reliable, and the paper well-written. Overall, the article provides significant additional support for the idea that the activity of PKM ζ is critical for the maintenance of hippocampal L-LTP and LTM. The article uses genetic methods, rather than simply pharmacological ones, to demonstrate that when PKM ζ is genetically deleted, PKC δ /λ, compensates for the missing PKC ζ .

Weaknesses:

The paper sets up what I believe is probably a false dichotomy between a structural explanation - a change in the number of synaptic connections among neurons - and the persistent kinase activity explanation for memory maintenance. Why are these two explanations necessarily antithetical? It is possible that an increase in synaptic connections and the ongoing activity of PKM ζ both contribute substantially to memory maintenance. The authors certainly don't provide any evidence that the number of synapses in the hippocampus remains unchanged after the induction of L-LTP or LTM. Indeed, I see no reason why persistent PKM ζ activity could not be a mechanism for the maintenance of an enhanced number of synaptic connections following the induction of LTP/LTM. To the best of

my knowledge, this possibility has not yet been explored. Consequently, I don't see why the present results would lead one to favor a biochemical explanation over a structural one for memory maintenance. Given the significant experimental evidence that LTM involves persistent structural changes in neurons, both explanations are equally plausible at present.

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Reviewer #2 (Public review):

Summary:

The authors are attempting to advance understanding of the role of unconventional PKCs, PKC ζ , and PKC ι/λ in maintenance of late-phase LTP. Their results help to clarify the interplay between "structural" and "biochemical/enzymatic" mechanisms of LTP and learning in the hippocampus.

Strengths:

A strength is the use of conditional knock-outs of PKC ζ and PKC ι/λ to assess the role of these two enzymes in maintaining long-term potentiation and in compensating for each other when one of them is conditionally knocked out in the adult.

Weaknesses:

The paper is extremely difficult to read because the abstract does not clearly state the advances made over earlier studies by the use of conditional KO mutation. For example, in line nine of the abstract, the authors state, "Here, we found PKC ι/λ persists in LTP and long-term memory when PKC ζ is genetically deleted." This is confusing because it sounds as though the experiments have repeated earlier published experiments in which the gene encoding PKC ζ is deleted in the embryo. The authors are not clear throughout the manuscript that they are using conditional KO of the two enzymes in the adult animal, rather than deletion of the gene. The term "genetically deleted" does not mean "conditionally deleted in the adult." The final sentences of the abstract are: "Whereas deleting PKC ζ and PKC ι/λ individually induces compensation, deleting both aPKCs abolishes hippocampal late-LTP. Hippocampal ι/λ - ζ -double-knockout eliminates spatial long-term memory but not short-term memory. Thus, in the absence of PKC ζ , a second persistent biochemical process compensates to maintain late-LTP and long-term memory." These sentences do not convey a clear logical conclusion. The Discussion does a better job of stating the importance of the experiments.

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Reviewer #3 (Public review):

Summary:

The manuscript addresses an important, yet unresolved and long-debated, question: whether atypical protein kinase C is required for the maintenance of late-long-term synaptic potentiation (L-LTP) and long-term memory (LTM). The authors confirm previous findings that persistent activity of PKC ζ is required for hippocampal L-LTP and spatial memory. They demonstrate that genetically deleting PKC ι/λ and PKC ζ individually induces compensatory upregulation, whereas deleting both atypical PKCs abolishes hippocampal L-LTP spatial long-term memory. The study uses an elegant combination of immunoblots, electrophysiology, and behavioral assays. The use of Cre-recombinase to target specific hippocampal regions and neurons adds to the rigor of the findings.

Strengths:

The manuscript addresses an important, yet unresolved and long-debated, question; whether PKM ζ is required for the maintenance of L-LTP and LTM. The study demonstrates that PKC ι/λ , which was previously shown to be critical for the initial generation of the early phase of LTP and short-term memory, becomes persistently active in L-LTP and LTM in a PKM ζ knock-out model, compensating for the loss of PKM ζ . Furthermore, when the compensation mechanisms are eliminated by simultaneous deletion of both PKM ζ and PKC ι/λ , maintenance of LTP and long-term spatial memory, but not of short-term memory, is diminished. The strength of this study is that the authors used a double-knockout strategy to directly address the controversy concerning the roles of PKM ζ in memory formation. By showing that PKC ι/λ compensates when PKM ζ is deleted, the authors provided a compelling explanation for previous contradictory findings.

Weaknesses:

- (1) The authors should provide the numerical values for all data.
- (2) It appears that blind procedures were only used for the behavioral experiments. Some explanation is warranted.
- (3) The description of the immunoblotting procedures lacks sufficient detail. The authors state that immunoblots were stained with multiple antisera to visualize multiple PKCs on the same immunoblot. To conserve antisera, the immunoblots were cut to isolate the relevant proteins based on molecular weight. Isoforms with similar molecular weights were either stained with antisera of different species or on separate blots. Despite this explanation, it is unclear how immunoblotting was performed in practice. For example, in Figure 1B, the authors compared the changes of four conventional PKC isoforms. Because all four antibodies are mouse monoclonal antibodies recognizing proteins of similar molecular weights, each probing should presumably have its own actin loading controls. However, these controls are missing from the figure. Some clarification is warranted.
- (4) The statement in the legend to Figure 4B, that the increases of maximum avoidance time from pretraining to trial 1 are not different, indicates both groups of mice successfully established short-term memory, which is not correct. The analysis only reveals that there is no difference between the two groups. No differences could be due to both groups learning the same, as the authors suggest, or alternatively to no learning in either group.
- (5) The labeling on some of the illustrations (e.g., Figure 2B) is unreadable.
- (6) In Figure 4B, only the single statistical comparison between "pretaining" and "1 trial" is shown. The other comparisons described in the legend should also be illustrated.
- (7) There is no documentation to support the statement that "The prevailing textbook mechanism for how memory is retained asserts that stable structural changes at synapses, the result of initial protein synthesis and growth, sustain memory without the need for ongoing biochemical activity dedicated to storing information" or for the statement in the Discussion that the structural model of memory storage is the standard account.

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Author response:

Thank you for the reviews of our article "PKM ζ -PKC ι/λ double-knockout demonstrates atypical PKC is crucial for the persistence of hippocampus LTP and spatial memory." We will address all of the reviewers' issues point-by-point in a revised version.

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